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<< THE GOOD EARTH

SUBHASH ARYA'S underground alternative to air-conditioning has fought cynicism and inertia and is now beating the heat

The office of the Commercial Taxes Department in Jaipur is the only building in the desert state of Rajasthan to have a earth tunnel cooling system. Constructed by the Rajasthan State Road Development Corporation (RSRDC), it does not use water or emit harmful gases as an air-conditioner does, and saves hugely on power. "The building's cutouts, tinted glasses and cooling tunnel combine to allow optimum use of sunlight, fresh flow of air, restrict transmission of heat and is a powersaver," says Subhash Arya, resident engineer with RSRDC, who got the concept from RETREAT (Resource efficient TERI Retreat for Environmental Awareness and Training) and executed it. Implemented in just one part of the building because of resistance to spending on innovations in the state, it has now become a pilot project for other buildings.

A Rs 10-lakh investment has resulted in cooling a 5,000-sq-ft portion of the building's ground floor—equal to cooling by a 20-tonne air-conditioner while consuming power equivalent to that of a four-tonne. For this, a tunnel was laid 14 ft below the ground which was connected to an airflow duct. This pipe sucks the air from outside, which passes through a 70m tunnel to come down to a temperature that is at least 15° C less than the temperature outside. So even in a city like Jaipur, it helps in maintaining a comfortable temperature of 28° C even when it's above 40° C outside. The continuous flow of fresh air at constant temperature not only kills germs, but also makes it odourless. The earth cooling system is a 'cool' way to reduce carbon emissions. **by Rohit Parihar**

<< GREEN DREAM

ANIL SEKHRI'S green house is an example of how eco-friendly not only means ecology-friendly but economy-friendly as well

Even in the middle of Mumbai's summers, Anil Sekhri's eco-friendly home does not require an air-conditioner. Terracotta clay tiles cover the ceilings of each room to ensure the house remains cool throughout the year. The 2,500-sq-ft home, which nestles in a green zone in suburban Mumbai, was built without cutting down a single tree. "The house is shaped around the trees. Too many people are cutting trees recklessly without realising that they absorb carbon dioxide and should be protected," he says. The chartered accountant's two-level home has wooden switchboards instead of plastic, and paint and plaster have not been used on the walls.

Keeping in mind the recent water shortage, Sekhri will also introduce water harvesting facilities for non-potable water supply for six months. The building materials, including Malad stone, were sourced locally to avoid transport pollution. Even beer bottles and furniture parts were picked up from scrapyards and fitted into the home. While most eco-friendly homes are equated with high cost, Sekhri completed the construction within a budget of Rs 25 lakh. The architects who designed the home were Tanya Mahajan and Malak Singh, who studied under British architect and pioneer of eco-friendly, low-cost homes in India Laurie Baker. Following his footsteps, Singh infused the British architecture into this home. "This home is truly a Baker style home, with little excesses. My friend M.F. Husain taught me that the more ostentatious a house, the more ugly it becomes," says Sekhri.

by Nishika Patel